William Castro

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EDUCATION

Boston University, College of Engineering — Boston, MA
Master of Science in Mechanical Engineering — *Candidate, December 2026*
Late Entry Accelerated Program (LEAP) — *September 2024 - Present*

University of Massachusetts Amherst, Isenberg School of Management — Amherst, MA
Bachelor of Business Administration in Operations and Information Management — *Graduated, May 2023*

- GPA 3.86; Dean's List All Semesters, Cum Laude Honors, Data Analytics Certificate

RESEARCH EXPERIENCE

Lubner Group, Boston University — Boston, MA
Direct Air Capture of CO_2 – Laboratory Assistant — *May 2026 - Present*

- Building hardware for the lab's planned 5,000-cycle degradation test of metal-organic frameworks (MOFs), porous materials that capture CO_2, where heating and cooling time per cycle sets how long the test takes
- Designed and CNC milled an aluminum vacuum chamber, with a waterjet-cut lid and laser-cut nitrile gasket
- Reached 0.8 mbar and traced leaks to the NPT fittings, not the chamber or gasket, using isopropanol testing; now switching to compression fittings to complete vacuum validation
- Designed and 3D printed an air funnel for a 120 mm fan, cutting tube cooldown from 160 °C to 35 °C from ~10 min to ~4 min (2.5x faster)
- Built and tested a second tube furnace core with thicker nichrome wire that reached 90 °C in under a minute, about twice as fast as the original
- Machined aluminum sample holders to ~0.0015" clearance in the tube bore, preventing powder blowout

PROJECTS

Rotary Vane Pump — *September 2026 - Present*

- Building a rotary vane pump in BU's Manufacturing Processes course using sand casting, CNC and manual machining, and drilling and tapping, working to the drawing specifications
- Bored the rotor hole to 0.7480" +0.0005/−0.0000, verifying it with a custom stepped plug gauge since no available tool could measure that tolerance, so the shaft could be shrink fit into it; the fit was successful
- Redesigning the pump with a team of 3 for liquid ammonia (−50 °F, 8 atm, 3,600 RPM, 4,500-hour MTBF), producing GD&T drawings to ASME Y14.5-2018

PressMate Coffee Press — *January 2026 - May 2026*

- Designed the PressMate, a steel coffee press, with a 4-person team in BU's Product Realization course, balancing a limited budget, safety, and customer use-case requirements
- Built 4 mockups and 14 steel prototypes (1 EVT, 3 DVT, 10 PVT) using CNC and manual lathes, waterjet cutter, TIG welding, cast silicone, and wire EDM in a 22-step process for each prototype
- Optimized the CNC lathe process for steel tube by testing inserts, boring bars, cut depths, speeds, feeds, and chuck clamping force, significantly improving surface finish
- Diagnosed plunger binding in the PVT as an O-ring groove that was too narrow, not surface finish, and fixed it with chamfers; every unit sealed and fit correctly
- Created and maintained 16 controlled documents, including RASCI and risk management, with templates and revision tracking through EVT, DVT, and PVT
- Developed a 10,000-unit mass production plan for the PressMate, including packaging plans, drawings, process plans, SOPs, tooling, and cost model

Automated Circuit Printing — *January 2026 - May 2026*

- Expanded a robotic cell of two CNC mills and three UR5 arms with a 6-person team, adding a potentiometer and buzzer to produce 2 functional circuits from milled HDPE and silver conductive ink
- Extended the Python control code, automated CAD/CAM, and pick-and-place programs for the new components; designed and milled an HDPE component tray
- Tuned UR5 pick-and-place in sub-millimeter increments so components landed accurately on the board
- Fixed uneven conductive-ink extrusion with a larger syringe tip; added a 2–3 mm code offset to protect the tool

Thermal Expansion Simulation of a Pump Housing *September 2025 - December 2025*

- Modeled a gray iron rotary vane pump housing in COMSOL (heat transfer coupled with solid mechanics) with a team of 3, building the geometry, constraints, and boundary conditions
- Predicted ~0.0015” bore growth from 70 °F to 140 °F and validated it against a hand calculation, showing thermal expansion was not significant for the pump design

Accumulator Cart for High Voltage Battery, Terrier Motorsports *November 2024 - February 2025*

- Led a 4-person team in designing a hydraulic accumulator cart to safely remove a high-voltage battery from a racecar, prioritizing ease of use and adaptability to future battery designs
- Incorporated a deadman brake, electrical insulation, and an integrated battery charger into the design to protect operators and improve usability
- Designed new wheel brackets to fit the brakes, validating a 500 lb load capacity with SolidWorks FEA

WORK EXPERIENCE

Ernst & Young (EY) Boston, MA

Client Serving Intern, Technology Risk Consulting *June 2022 - August 2022*

- Completed SOX control testing for various clients, including change and access controls, to ensure compliance
- Documented audit walkthroughs and testing results to support audit readiness and adherence to standards
- Supported engagement teams with project planning and client deliverables

Launch Intern *June 2021 - August 2021*

- Tested controls to ensure design and operating effectiveness
- Rotated between consulting, audit, and tax client-serving teams
- Worked with fellow interns on a capstone project to help solve a Tennessee labor problem

Miranda Bread Marlborough, MA

Operations and Maintenance Coordinator *Summer 2017 - January 2023*

- Maintained all company vehicles, including two Sprinter vans
- Handled day-to-day repairs and replacements of broken equipment to keep each bakery running
- Troubleshot network, POS system, printer, and office computer issues across locations
- Managed inventory deliveries to store locations and tracked inventory across locations and a production plant
- Reconciled receipts for money transfer services worth over $100,000 in a single shift
- Monitored store cleanliness and employee conduct across locations

SKILLS

Engineering Software: SolidWorks, Fusion 360, Onshape, COMSOL, FEA, Python, MATLAB
Design Standards & Methods: GD&T (ASME Y14.5-2018), DFM/DFA, Root Cause Analysis, Sealing Design
Data & Business Tools: SQL, R, JMP, Tableau, Power BI, VBA, SAP, Microsoft Excel
Manufacturing: CNC Mill, CNC Lathe, Manual Mill, Manual Lathe, Waterjet Cutter, Laser Cutter, 3D Printing, TIG Welding, Sand Casting, Soldering
Language: Conversational in Portuguese

LEADERSHIP & ACTIVITIES

Operations & Information Management Club, Association of Latino Professionals for America

Treasurer, Fundraising Chair *September 2019 - May 2023*

- Managed club finances, including the bank and Venmo accounts, and created yearly budgets
- Organized fundraisers to make sure the club had the funds to operate
- Connected with recruiters, professionals, and alumni in the operations and information management field